

Bounded Increments and Rational Reciprocal Sums

A sufficient condition for the Sylvester recurrence in Erdős Problem #243

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ABSTRACT

Let $a_1 < a_2 < \dots$ be positive integers with $a_{n+1}/a_n^2 \rightarrow 1$ and $\sum_n 1/a_n \in \mathbb{Q}$. Put $P_n = \prod_{j < n} a_j$. We prove that an eventual upper bound on $P_{n+1}/a_{n+1} - P_n/a_n$ forces $a_{n+1} = a_n^2 - a_n + 1$ eventually. This strengthens a sufficient condition requiring the upper limit of these increments to be nonpositive. Under this bound, clearing denominators in the rational tails produces positive integer numerators with bounded upward increments. Integer descent handles an eventually nonnegative error; otherwise, stabilising the common divisor makes a Chinese remainder theorem obstruction available. The needed increment bound is not derived from growth and rationality alone, so this sufficient condition does not resolve Erdős Problem #243.

1 Introduction

For integers $a_n > 1$, the Sylvester recurrence $a_{n+1} = a_n^2 - a_n + 1$ gives the telescoping identity

$$\frac{1}{a_n - 1} = \frac{1}{a_n} + \frac{1}{a_{n+1} - 1}.$$

We call a sequence that satisfies this recurrence eventually a *Sylvester tail*. Its reciprocal sum from any index n in that tail onwards is $1/(a_n - 1)$. Erdős Problem #243 asks whether rationality and $a_{n+1} \sim a_n^2$ force this recurrence. We prove the following sufficient condition.

Corollary 1.1 (bounded increments of P_n/a_n). *Let $a_1 < a_2 < \dots$ be positive integers, $a_{n+1}/a_n^2 \rightarrow 1$, and $\sum_{n \geq 1} 1/a_n \in \mathbb{Q}$. Put $P_n = \prod_{j < n} a_j$. If*

$$\limsup_{n \rightarrow \infty} \frac{P_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < +\infty,$$

then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n .

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The assumption bounds the upward increments of P_n/a_n ; it does not bound P_n/a_n itself. Indeed,

$$\frac{P_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) = \frac{P_{n+1}}{a_{n+1}} - \frac{P_n}{a_n}.$$

The hypothesis is one-sided: it imposes no lower bound on the increments and does not require monotonicity. The growth hypothesis alone says that the ratio of consecutive terms of P_n/a_n tends to one; it does not bound their difference.

For a Sylvester sequence starting at $a_1 > 1$, $a_n - 1 = (a_1 - 1)P_n$, so the increments tend to zero. For $a_n = b^{2^{n-1}}$, with integer $b \geq 2$, they are identically zero; since this sequence has no Sylvester tail, the corollary recovers the irrationality of its reciprocal sum. The bound also allows positive limiting increments: the integer sequence $a_1 = 4$, $a_{n+1} = \lceil na_n^2/(n+1) \rceil$, beginning 4, 8, 43, 1387, ..., is such an example. We verify this after the main proof in Section 4. A ratio tending to one alone does not give the bound: the rounded cubic-rate example in Section 7 has increments of order n^2 .

Write the rational tail as C_n/D_n , clearing denominators without reducing at each step. Up to a finite shift and a common positive factor, these are Koizumi's tail numerators and denominators [10, Corollary 3, p. 9; Lemma 4, pp. 11–12]. The error $E_n = D_n - (a_n - 1)C_n$ measures departure from the Sylvester tail, and $C_{n+1} = C_n - E_n$ turns a lower error bound into an upper increment bound. If the error does not vanish eventually, the numerator tends to infinity and negative errors occur arbitrarily late. Each gcd divides every later error, so the bounded negative magnitudes force the gcd to stabilise. Dividing it out gives coprimality to every earlier multiplier. The Chinese remainder theorem then supplies a forbidden block that bounded upward jumps cannot cross without landing inside.

Here is the precise arithmetic statement. For the rational tails, Section 4 supplies every condition except (5). Condition (4) follows from (6) with $K = 1$; listing it separately makes the hypothesis used when propagating a zero error explicit.

Theorem 1.2 (bounded negative part). *Let $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$ and $E : \mathbb{N} \rightarrow \mathbb{Z}$ satisfy*

1. $a_n > 1$ and $C_n > 0$ for every n ;
2. the exact dynamics $C_{n+1} + D_n = a_n C_n$ and $D_{n+1} = a_n D_n$;
3. $E_n = D_n - (a_n - 1)C_n$ for every n ;
4. eventual strict centring: $|E_n| < C_n$ for all large n ;
5. eventually bounded negative part: $-B \leq E_n$ for all large n , for some integer $B \geq 0$;
6. vanishing relative error: for every integer $K \geq 1$ there is an N with $K|E_n| < C_n$ for all $n \geq N$.

Then $E_n = 0$ for all sufficiently large n .

Condition (5) bounds upward steps but permits sign changes and places no independent upper bound on positive errors. Condition (6) is $|E_n|/C_n \rightarrow 0$. Absorption needs only the weaker centring bound (4), not $-C_n/2 \leq E_n < C_n/2$. The scalar example $C_n = n + 1$, $E_n = -1$ satisfies the update and conditions (5)–(6) without stabilising: the relations involving the denominator are essential.

Problem 1.3 (Erdős #243). Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n .

The question is recorded by Erdős and Graham [5] and by Erdős [6, p. 105], and Bloom's catalogue lists it as Problem #243 [11].

Sections 2–4 contain the main proof: the arithmetic argument, its gcd and CRT lemmas, and the construction from the reciprocal series. The later sections give extensions and state the estimate still needed for the unrestricted problem. Appendix A records the formal correspondences separately.

2 Proof under a lower bound on the error

We first assemble the argument. Section 3 proves its arithmetic lemmas; Section 4 constructs the integers from the rational reciprocal sum.

Proof of Theorem 1.2. Suppose the error is not eventually zero. By Theorem 3.2, after the centring threshold a single zero would force every later error to be zero. Hence $|E_n| \geq 1$ on a sufficiently late tail, and condition (6) gives $C_n \rightarrow \infty$.

If there were only finitely many negative indices, Theorem 3.2 would give eventual zero. There are therefore infinitely many, and condition (5) bounds their magnitudes by B . In particular $B \geq 1$. Proposition 3.7 now makes $G_n = \gcd(C_n, D_n)$ constant, say $g > 0$, from some index onwards. On this tail $u_n = C_n/g$ and $v_n = D_n/g$ form a reduced exact orbit, with $u_n \rightarrow \infty$ and

$$u_{n+1} - u_n = -E_n/g \leq B.$$

The earlier multipliers are coprime to every later reduced numerator. Theorem 3.5 says that these coprimality conditions are incompatible with divergence and the displayed bound on upward steps. $\square$

Corollary 2.1. *Under Theorem 1.2, the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ eventually.*

Proof. Theorem 1.2 gives eventual zero error. Since $C_n > 0$, Corollary 3.3 gives the recurrence. $\square$

3 The arithmetic ingredients

We write $z_+ = \max(z, 0)$ and take empty products and least common multiples to be 1. An *exact state* means sequences satisfying the two recurrences and the definition of E_n in Theorem 1.2; positivity and bounds on the error are stated separately. Removing finitely many terms does not change convergence or rationality of the reciprocal series, or the eventual validity of a recurrence. A specified higher-order rate retains its original index: reindexing changes its lower-order terms.

3.1 Error, absorption and descent

The error identity gives both the Sylvester recurrence and propagation of a zero error. Descent then handles an eventually nonnegative error.

Proposition 3.1 (error identities). *For an exact integer state,*

$$C_{n+1} = C_n - E_n, \quad (a_{n+1} - a_n^2 + a_n - 1)C_{n+1} = a_n^2 E_n - E_{n+1}.$$

Proof. Substitute $E_n = D_n - (a_n - 1)C_n$ and use $D_{n+1} = a_n D_n$, $C_{n+1} = a_n C_n - D_n$. $\square$

Theorem 3.2 (absorption and descent). *For a positive exact state with strict centring, $E_n = 0$ implies $E_{n+1} = 0$. For any positive integer state with $C_{n+1} = C_n - E_n$, eventual nonnegativity of E_n implies its eventual vanishing.*

Proof. When $E_n = 0$, the second identity makes E_{n+1} a multiple of C_{n+1} ; strict centring forces that multiple to be zero. In the second assertion, C_n is eventually a nonincreasing sequence of positive integers, so it stabilises. $\square$

Corollary 3.3 (two zero errors). *If $E_n = E_{n+1} = 0$ and $C_{n+1} \neq 0$, then $a_{n+1} = a_n^2 - a_n + 1$. Thus eventual zero error in a positive exact state implies the eventual Sylvester recurrence.*

Proof. The second error identity has nonzero factor C_{n+1} . $\square$

Absorption uses strict centring at the successor index. After an eventual centring threshold, a zero therefore propagates to every later index. On a tail that is not eventually Sylvester, no zero can occur beyond that threshold. Then $|E_n| \geq 1$, so vanishing relative error gives $C_n \rightarrow \infty$.

3.2 The first-crossing obstruction

Lemma 3.4 (consecutive multiples). *For pairwise coprime integers $m_0, \dots, m_{B-1} \geq 2$ and every lower bound, there is a larger t such that $m_i \mid t + i$ for each $i < B$.*

Proof. Solve $t \equiv -i \pmod{m_i}$ by the Chinese remainder theorem, then add multiples of $\prod_i m_i$. $\square$

Lemma 3.4 matches each modulus m_i to its own multiple $t + i$ inside a window of B consecutive integers. The same forbidden block and first-crossing contradiction, under a two-sided bound on the error, appear in the proof of Bado's Theorem 5.1 [7, pp. 4–5]. The lemma here isolates bounded upward movement, and the gcd stabilisation below makes it applicable with only a lower bound on the error.

For example, the consecutive integers $6k + 2, 6k + 3$ are forbidden to a sequence coprime to both 2 and 3. Upward jumps of size at most 2 cannot cross this pair from below without landing in it. The following statement is the same first-crossing argument with a block supplied by the Chinese remainder theorem.

Theorem 3.5 (Chinese remainder theorem and first crossing). *Let $u : \mathbb{N} \rightarrow \mathbb{N}$ tend to infinity and let $B \geq 1$ be an integer with $u_{n+1} \leq u_n + B$ for every n . There is no sequence of pairwise coprime integers $m_i \geq 2$ for which $\gcd(m_i, u_i) = 1$ whenever $i < t$.*

Proof. Choose $m_0, \dots, m_{B-1}$ and use Lemma 3.4 to find $t > \max(u_0, \dots, u_B)$ with $m_i \mid t + i$. Since $u_n \rightarrow \infty$, there is a first $n > B$ with $u_n \geq t$. Minimality and the rise bound give

$$t \leq u_n \leq u_{n-1} + B < t + B.$$

Hence $u_n = t + i$ for some $0 \leq i < B < n$, so $m_i \mid u_n$. This contradicts $\gcd(m_i, u_n) = 1$. The argument uses first crossing, not monotonicity: the numerator may fall before it reaches the block. $\square$

Only unboundedness above is needed for the first crossing. The stated hypothesis $u_n \rightarrow \infty$ holds in the application to reduced tails.

3.3 Reduction and stabilisation

A reduced exact tail has positive u_n , integer $v_n \geq 0$, $\gcd(u_n, v_n) = 1$, and

$$u_{n+1} = a_n u_n - v_n, \quad v_{n+1} = a_n v_n.$$

Proposition 3.6 (persistent coprimality). *In a reduced exact tail, $\gcd(a_n, v_n) = 1$. Distinct multipliers are pairwise coprime, and every earlier multiplier is coprime to every later numerator.*

Proof. A common prime divisor of a_n, v_n would divide both u_{n+1} and v_{n+1} . Also, $a_i \mid v_t$ for $i < t$, so reducedness gives $\gcd(a_i, u_t) = 1$, and $\gcd(a_t, v_t) = 1$ gives $\gcd(a_i, a_t) = 1$. $\square$

Proposition 3.7 (gcd stabilisation). *For a positive exact state, suppose that some fixed integer $B \geq 1$ satisfies $-B \leq E_n < 0$ at infinitely many indices. Then $G_n = \gcd(C_n, D_n)$ is eventually constant. Division by its stable value gives a reduced exact tail.*

Proof. Both updates preserve common divisors, so $G_n \mid G_{n+1}$, and $G_n \mid E_n$. For any n , choose $t \geq n$ with $-B \leq E_t < 0$. Then $G_n \leq G_t \leq -E_t \leq B$. Thus the entire positive divisibility chain is bounded and eventually constant, even if other negative errors are larger. Division by its stable value preserves both exact updates and leaves the states coprime. $\square$

4 Constructing the integers from a reciprocal series

Assume the positivity, strict increase, growth and rationality hypotheses of Corollary 1.1, but not yet its increment bound. We use the denominator-clearing construction of [10, Lemma 4, pp. 11–12], with the following indexing. Let $x_n = \sum_{k \geq n} 1/a_k$. Write $x_1 = p/q$ with positive integers p, q , and put

$$P_n = \prod_{j < n} a_j, \quad D_n = qP_n, \quad C_n = D_n x_n.$$

Each C_n is a positive integer: explicitly,

$$C_n = pP_n - q \sum_{k < n} P_n / a_k.$$

Every quotient in the finite sum is an integer. Positivity follows from $x_n > 0$. The identity $x_n = 1/a_n + x_{n+1}$ now gives both exact updates; no approximation or rounding occurs in this construction.

For all large n , $a_{n+1} \geq a_n^2/2 \geq 2a_n$. Thus the terms after $1/a_{n+1}$ sum to at most $2/a_{n+2} \leq 4/a_{n+1}^2$, and

$$x_n = \frac{1}{a_n} + \frac{1}{a_{n+1}} + O(a_{n+1}^{-2}).$$

Consequently

$$\frac{C_{n+1}}{C_n} = \frac{a_n x_{n+1}}{x_n} = \frac{a_n^2}{a_{n+1}} + O(1/a_n) \rightarrow 1, \quad \frac{E_n}{C_n} \rightarrow 0.$$

In particular, $|E_n| < C_n$ eventually. Koizumi obtains the relative-error limit for the gap of the eventual pseudo-greedy expansion [10, Corollary 3, p. 9]. Under the common rescaling of his integers c_n, d_n, e_n to C_n, D_n, E_n [10, pp. 11–12], that gap is E_n/C_n ; his rounding rule gives $-C_n/2 \leq E_n < C_n/2$ on that tail.

Proof of Corollary 1.1. Only a one-sided comparison is needed. Substituting $x_n = 1/a_n + 1/a_{n+1} + \sum_{k \geq n+2} 1/a_k$ gives the exact identity

$$E_n + q \frac{P_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) = q P_n \left(\frac{1}{a_{n+1}} - (a_n - 1) \sum_{k \geq n+2} \frac{1}{a_k} \right). \quad (1)$$

The tail bound makes the subtracted term at most $4(a_n - 1)/a_{n+1}^2 < 1/a_{n+1}$ eventually, since $a_{n+1}/a_n \rightarrow \infty$. The right side is therefore positive. The assumed upper bound on the product expression now supplies a lower bound on E_n . After deleting a finite prefix, Theorem 1.2 and Corollary 2.1 give the result. $\square$

This completes the main argument. The summability, LCM and higher-order rate hypotheses in the following extensions are not used in it.

The product-weighted comparison in (1) also occurs in Koizumi's proof of his Corollary 4(1) [10, (11)–(12), p. 15]. Here its eventual sign suffices for the lower error bound.

An example with positive limiting increments. For $a_1 = 4$ and $a_{n+1} = \lceil na_n^2/(n+1) \rceil$, $a_{n+1} \geq a_n^2/2$ gives $a_n \geq 2 \cdot 2^{2^{n-1}}$, so the sequence is strictly increasing. The rounding remainder lies in $[0, 1)$, hence

$$0 \leq 1 + \frac{1}{n} - \frac{a_n^2}{a_{n+1}} < \frac{4}{a_n^2}.$$

Dividing the consecutive ratio of P_n/a_n by $(n+1)/n$ leaves positive factors with absolutely summable deviations from one. Thus $P_n/a_n \sim Kn$ for some $K > 0$. Since $a_n^2/a_{n+1} - 1 \sim 1/n$, the increments, equal to the product of these two quantities, tend to K . The corollary proves irrationality, since a Sylvester tail would instead have $a_n^2/a_{n+1} - 1 = O(1/a_n)$. The positive limiting increment separates the two product conditions, not the full range of earlier irrationality criteria.

The additional hypothesis in Theorem 1.2 is the lower bound on E_n . The growth and rationality assumptions already supply the other arithmetic hypotheses after a finite

shift. Badea's criterion [3, Corollary 2.2, p. 316] assumes instead the eventual growth inequality $a_{n+1} \geq a_n^2 - a_n + 1$ and forces equality under rationality. Koizumi's nonnegative-error result is integer descent [10, Proposition 1(2), p. 14]. These are not the same one-step inequality, although under the standing hypotheses either one, imposed eventually, forces a Sylvester tail and hence the other. The present theorem allows sign changes but bounds the magnitude of negative errors. Koizumi's Corollary 4(1) assumes a nonpositive upper limit for the product expression [10, pp. 14–15], and Corollary 1.1 allows any finite upper limit. Bado's preprint, posted in September 2026, assumes two-sided bounded errors [7, Theorem 5.1, p. 4]; Theorem 1.2 needs only an eventual lower bound. Both proofs use a forbidden block from the Chinese remainder theorem. Here the one-sided bound first stabilises the gcd, which makes that argument available for the reduced numerators.

The error identities, absorption and descent follow from Koizumi's recurrences [10, Lemmas 2–4, pp. 10–12]; he credits the nonnegative-error case to Badea [3, Corollary 2.2, p. 316]. Earlier remainder methods are discussed in Appendix A.

5 Finite total relative increase

The next condition concerns the sum of all upward increments of C_n , each divided by the value before that step. A Sylvester tail has no such increments after a finite index. In contrast, $C_n = n + 1$ and $C_n = (n + 1)^2$ both have divergent sums, although their relative increments tend to zero. Summability is thus much stronger than that limit. The denominator recurrence is not needed for the following argument.

Theorem 5.1 (a convergent sum of relative increases). *Let C_n be positive integers and E_n integers satisfying $C_{n+1} = C_n - E_n$. If*

$$\sum_n \frac{(-E_n)_+}{C_n} < \infty,$$

then $E_n = 0$ eventually. Neither denominator dynamics nor vanishing relative error is required.

Proof. Set $\delta_n = (-E_n)_+/C_n$. Then

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n) \leq C_0 \exp\left(\sum_n \delta_n\right).$$

Choose an integer upper bound K for C_n . Each strict rise contributes at least $1/K$ to $\sum \delta_n$, so there are only finitely many rises. The remaining positive integer sequence is nonincreasing and stabilises, and the update then gives $E_n = 0$. $\square$

The proof also allows positive real C_n with integer E_n : the values lie in $C_0 + \mathbb{Z}$, whose bounded positive part is finite. Discrete increments are essential to this argument. With real errors, $C_n = 1 + 1/(n+1)$ decreases forever with zero relative-increase sum; without positivity, $C_n = -n - 1$, $E_n = 1$ does so too. The companion's Section 13 gives the details.

On an exact reciprocal-tail orbit the conclusion gives the Sylvester recurrence. For the gap sequence of the pseudo-greedy expansion, the same criterion, with the same

product bound and integer descent, appears in the Erdős Problem a Day working report on Problem #243, dated 12 August 2026 [8, “A global termination criterion”]; the statement above is for any positive integer sequence with $C_{n+1} = C_n - E_n$. This summability condition and the bound in Theorem 1.2 are different estimates. For exact reciprocal tails with $|E_n|/C_n \rightarrow 0$, each is equivalent to eventual vanishing of E_n ; neither estimate has been derived here from the unrestricted problem.

6 A criterion using new maxima of an LCM numerator

Instead of multiplying all earlier denominators, one can clear the rational tail with their least common multiple. We will sum only at steps where the resulting numerator exceeds all its earlier values. This allows some increases to be omitted from the sum and others to receive smaller weights. The proof is given below. The separate release contains [the real-weight criterion](#); it is outside the checked main-repository build documented here. Set

$$L_n = \text{lcm}(q, a_1, \dots, a_{n-1}), \quad M_n = D_n/L_n, \quad U_n = C_n/M_n, \quad V_n = E_n/M_n.$$

The rational tail has denominator dividing L_n , so U_n and V_n are integers. This need not be a reduced fraction: U_n and L_n may share a factor. With $\rho_n = \gcd(L_n, a_n)$, the exact updates are

$$M_{n+1} = M_n \rho_n, \quad \rho_n U_{n+1} = U_n - V_n, \quad V_n = L_n - (a_n - 1)U_n.$$

These coordinates and updates also appear in Bado’s September 2026 preprint. With his denominator parameter equal to q , his M_{n-1} , Δ_{n-1} , K_{n-1} , u_n and g_n are respectively our L_n , M_n , U_n , V_n and ρ_n ; his (19) is the middle update above [7, Proposition 7.1 and (16)–(20), pp. 6–7].

Since $L_n \mid qP_n$ and $V_n = (L_n/(qP_n))E_n$, multiplying (1) by $L_n/(qP_n)$ gives, eventually,

$$0 < V_n + \frac{L_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) \leq \frac{L_n}{a_{n+1}}. \quad (2)$$

Its lower bound gives the LCM corollary below. Summability of the comparison error is proved when needed for the weighted criterion; the different denominator update is accounted for by ρ_n .

Write $R_n = \max_{j \leq n} U_j$ and $\mathcal{R} = \{n : U_{n+1} > R_n\}$. Strict centring already gives $U_{n+1} < U_n$ when $\rho_n \geq 2$, so every sufficiently late strict rise has $\rho_n = 1$. The stronger eventual bound $-U_n \leq 2V_n$, supplied by $V_n/U_n = E_n/C_n \rightarrow 0$, gives the quantitative estimate $U_{n+1} \leq 3U_n/4$ when $\rho_n \geq 2$. At a sufficiently late record step, where $\rho_n = 1$, the actual jump is $d_n = U_{n+1} - U_n = -V_n > 0$. This identity is not asserted at a contracting step with $\rho_n \geq 2$.

Theorem 6.1 (a convergent weighted sum over new maxima). *Assume the growth and rationality hypotheses of Problem 1.3. Let $f : [1, \infty) \rightarrow [0, \infty)$ be finite and nonincreasing, with $\int_1^\infty f(t) dt = \infty$. Then the sequence is eventually Sylvester if and only if, for some integer $B \geq 0$,*

$$\sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

Only a step reaching a new maximum contributes. At a late such step, $-V_n = U_{n+1} - U_n$, so the first B units of the increase are ignored. A Sylvester tail makes the sum finite because $V_n = 0$ eventually. On a non-Sylvester tail, the proof gives divergence for every admissible f and every fixed B . The limit $V_n/U_n \rightarrow 0$ gives a pointwise relative-error estimate; no bound on this weighted sum is derived from that limit here. The restriction on f matters: $f(t) = 1/t$ is permitted, as is $1/[t \log(et)]$, but $1/t^2$ is not. For the latter weight, the integral in the crossing argument is finite and cannot contradict unbounded growth.

Proof. Suppose the sequence is not eventually Sylvester. Absorption and integrality give

$$\frac{1}{U_n} \leq \frac{|V_n|}{U_n} = \frac{|E_n|}{C_n} \rightarrow 0.$$

Thus $U_n \rightarrow \infty$, so there are infinitely many record steps. At every sufficiently late one, $\rho_n = 1$, so a_n is coprime to L_n . The corresponding terms a_n are pairwise coprime: an earlier term divides the later L_n , while the term at that record is coprime to L_n . For any fixed $B \geq 1$, choose B such multipliers $m_0, \dots, m_{B-1} > B$ and take T after their indices and after the threshold beyond which records have $\rho_n = 1$. Their size follows from $a_n \rightarrow \infty$.

Put $P = \prod_i m_i$ and choose x by the Chinese remainder theorem with $m_i \mid x + i$. Consider all translates $\tau = x + B + kP > R_T, k \in \mathbb{Z}$; the first is at most $R_T + P$. We first show that crossing one of these heights requires a jump larger than B , then count how many heights a single jump can cross. Here $d_n = U_{n+1} - U_n$ is the actual jump, including any part that recovers an earlier decrease. A first crossing $U_n \leq R_n < \tau \leq U_n + d_n$ is a record step. If $d_n \leq B$, then $U_n \in [\tau - B, \tau)$, so some m_i divides U_n . It also divides L_n , hence divides $d_n = (a_n - 1)U_n - L_n$, contradicting $0 < d_n \leq B < m_i$.

If the step first crosses $h \geq 1$ such heights, their spacing gives $(h - 1)P < d_n$. With $r = d_n - B \geq 1$ and $P \geq B + 1$, we have $d_n = B + r \leq Pr$, whence $h \leq r$. Monotonicity of f now gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \leq (d_n - B)f(U_n).$$

Each selected height above R_T has exactly one first crossing. The first selected height is at most $R_T + P$; each later height is P further on. For $R_N \geq R_T + P$, comparison of each interval with its left endpoint gives

$$\sum_{\substack{T \leq n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \geq \frac{1}{P} \int_{R_T+P}^{R_N} f(t) dt. \quad (3)$$

Since $R_n \rightarrow \infty$, the right side diverges for every $B \geq 1$; $B = 0$ follows by domination. Conversely a Sylvester tail telescopes to $x_n = 1/(a_n - 1)$, so $V_n = 0$ eventually. $\square$

For example, $f(t) = 1/[t \log(et)]$ gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (3) is $P^{-1} \log(\log(eR_N)/\log(e(R_T+P)))$. Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. These are specialisations of one crossing theorem.

The criterion also has an exact expression in the original growth defect. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $\theta_n = E_n/C_n$. The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \quad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Thus the two nonnegative summands $U_n f(U_n)(\gamma_n - B/U_n)_+$ and $(-V_n - B)_+ f(U_n)$ differ by at most $3U_n f(U_n)/a_n$. Here summability, not merely a one-sided bound, is needed. Since $a_n x_n \rightarrow 1$, we have $C_n/a_n \sim qP_n/a_n^2$, and

$$\frac{P_{n+1}/a_{n+1}^2}{P_n/a_n^2} = \frac{a_n^3}{a_{n+1}^2} \rightarrow 0.$$

The ratio test gives $\sum_n C_n/a_n < \infty$; now $U_n \leq C_n$ and $f(U_n) \leq f(1)$ make the comparison error summable. The errors in (1) and (2) are also absolutely summable, since each is eventually between zero and $qP_n/a_{n+1} = O(P_n/a_n^2)$. Consequently Theorem 6.1 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_n f(U_n) \left(\frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \quad (4)$$

for some B . The original hypotheses do not currently supply this finiteness. In particular, termwise convergence to zero is insufficient. Also, d_n is the actual jump, including any recovery from a drawdown; it must not be replaced by $R_{n+1} - R_n$ in the crossing proof.

Corollary 6.2 (a bound using the least common multiple). *Assume the hypotheses of Problem 1.3. Write $A_n = \text{lcm}(a_1, \dots, a_{n-1})$ with $A_1 = 1$. If*

$$\limsup_{n \rightarrow \infty} \frac{A_n}{a_n} \left(\frac{a_n^2}{a_{n+1}} - 1 \right) < \infty,$$

then the sequence is eventually Sylvester.

Proof. Since $L_n/A_n = q/\gcd(q, A_n)$ lies in $[1, q]$, an upper bound on the expression in the corollary gives an upper bound on $(L_n/a_n)(a_n^2/a_{n+1} - 1)$; negative terms remain negative. Equation (2) therefore bounds the negative part of V_n . Choose an integer B above that eventual bound. The record series then vanishes after a finite prefix, so Theorem 6.1 applies. $\square$

Erdős and Straus assume a nonpositive upper limit in this LCM expression [1, Theorem 3, p. 132]: their N_k is A_{k+1} and their growth ratio has index $k+1$. Tijdeman and Yuan extend this type of criterion to positive numerators [4]. Here any finite upper bound suffices under the quadratic-limit assumption. Since $A_n \mid P_n$, the LCM hypothesis is no stronger than the product hypothesis in Corollary 1.1: if the product expression is at most B , the LCM expression is at most $\max(B, 0)$. When the earlier terms are pairwise

coprime the two weights agree; repeated prime factors can make the LCM much smaller. This comparison does not assert the existence of a non-Sylvester rational example satisfying one bound but not the other.

Lean checks Corollary 6.2 as [the LCM condition for the reciprocal sequence](#). That declaration takes the rational reciprocal sum and the quadratic growth limit as hypotheses, with the finite upper limsup recorded as an eventual upper bound on the expression in the corollary. It does not assume that q divides the least common multiple of the earlier a_j .

For comparison, the all-positive, absolute-convergence specialisation of Duverney's signed-series criterion [2, Corollary 3.2, p. 287] already satisfies the classical product condition: $\sum_n |a_{n+1}/a_n^2 - 1| < \infty$ makes P_n/a_n converge to a positive finite limit, so its increments tend to zero. The signed recurrence and the distinction between this sufficient hypothesis and Duverney's printed signed-convergence display are set out in the companion's Section 3. The qualification concerns the nonzero infinite product used in the proof, not a claimed counterexample to Duverney's arithmetic theorem. Duverney, Kurosawa and Shiokawa [14, Theorem 1, author-version p. 2] address irrationality exponents under additional growth assumptions, including $x_{n+1} \geq x_n^2$ eventually; that inequality excludes a Sylvester tail. The fuller comparison is also in the companion's Section 3.

For integer summand numerators b_n , the updates become $V_n = b_n L_n - (a_n - 1)U_n$ and $\rho_n U_{n+1} = U_n - V_n$. With positive integers $L_n, U_n, a_n \geq 2$ and $L_{n+1} = \text{lcm}(L_n, a_n)$, an eventual lower bound on V_n already bounds U_n ; the additional limit $V_n/U_n \rightarrow 0$ gives eventual constancy. Section 6 of the companion, under "Integer coefficients", proves these two assertions separately, without assuming centring for the boundedness assertion. For positive b_n , compare Badea [3, p. 316] and Tijdeman–Yuan [4]. The pinned [coefficient proof supplement](#) also gives finite examples separating boundedness from stationarity.

7 Further consequences

Corollary 7.1 (an inclusive one-sided $1/n$ bound). *Let $a_1 < a_2 < \dots$ be positive integers with $a_{n+1}/a_n^2 \rightarrow 1$ and $\sum_n 1/a_n \in \mathbb{Q}$. Suppose that for some $K \geq 0$ and $\varepsilon > 0$,*

$$\gamma_n := a_n^2/a_{n+1} - 1 \leq \frac{1}{n} + \frac{K}{n^{1+\varepsilon}} \quad \text{for all large } n.$$

Then the sequence is eventually Sylvester. In particular, the conclusion holds under the pointwise eventual bound $\gamma_n \leq 1/n$.

Proof. Set $t_n = P_n/a_n$, so $t_{n+1}/t_n = 1 + \gamma_n$. For all large n ,

$$1 + \gamma_n \leq (1 + 1/n)(1 + K/n^{1+\varepsilon}).$$

The first product telescopes and the second is bounded, giving $t_n = O(n)$. Since $(\gamma_n)_+ = O(1/n)$, the increment $t_n \gamma_n$ is bounded above. This supplies the extra hypothesis of Corollary 1.1. $\square$

This is a written consequence of the bounded-increment criterion. It does not assert that the weaker condition $\limsup n(\gamma_n)_+ \leq 1$ suffices; a vanishing but nonsummable excess over $1/n$ is not covered by this proof.

Theorem 7.2 (cubic-rate irrationality). *A strictly increasing sequence of positive integers with*

$$a_n^2/a_{n+1} = 1 + \frac{3}{n} + o(n^{-3})$$

has irrational reciprocal sum.

Proof by the polynomial exclusion. Under rationality, Section 4 gives $C_{n+1}/C_n = 1 + 3/n + o(n^{-3})$. Integer finite differences then give $C_n = An(n+1)(n+2) + B$ eventually, with $A \in \mathbb{Q}_{>0}$ and $B \in \mathbb{Q}$. The fixed-cubic exclusion contradicts this profile. The complete ordinary argument, including the finite-difference extraction and the cubic-field step, is in Section 2 of the [companion reasoning paper](#). The field step uses Chebotarev; see [9, Section 3, author version] for that classical input. This paragraph is a proof by a stated companion result, not a standalone finite-congruence proof and not an assembled Lean proof. $\square$

For example, $a_1 = 8$ and $a_{n+1} = \lceil na_n^2/(n+3) \rceil$ give this rate and product-ratio increments of order n^2 . Thus the bounded-increment criterion does not apply, but Theorem 7.2 gives irrationality. The rounding estimates are in the companion's Section 2; its Section 7 treats the general $1 + c/n$ comparison.

The same finite-difference extraction excludes every nonintegral $\lambda > 1$ under the rate $a_n^2/a_{n+1} = 1 + \lambda/n + o(n^{-\lambda})$: eventual polynomial growth would force its degree to equal λ . The extraction uses $\Gamma(n+\lambda)/\Gamma(n) \sim n^\lambda$ [17, 5.11.12]; its full proof is in the companion paper. These rates fall outside $1 + o(1/n)$, the rate at which Koizumi notes that the Erdős–Straus criterion settles the problem [10, Remark 3, p. 16], and they make Duverney's signed series $\sum_n (a_{n+1}/a_n^2 - 1)$ diverge.

Although C_n is an integer linear form in the reciprocal sum, it grows. The argument uses an exact polynomial formula, not a small-linear-form criterion.

There is also an ordinary quantitative extension, not an end-to-end Lean-checked theorem. Under the hypotheses of Theorem 1.2 other than (5), the companion proves

$$\limsup_{n \rightarrow \infty} \frac{(-E_n)_+}{\log_2 \log_2 \max(4, C_n)} \leq 1 \quad \Rightarrow \quad E_n = 0 \text{ eventually.}$$

The companion's Theorem 7.3 and Corollary 7.4 prove this inclusive bound by controlling the height of a CRT block. A fixed bound satisfies it on a nonzero tail, where $C_n \rightarrow \infty$; a bound of order $\sqrt{C_n}$ need not. Unlike boundedness, the coefficient depends on the integer normalisation: scaling (C_n, D_n, E_n) by $k \in \mathbb{N}_{>0}$ scales this limit superior by k . The proof and the running-maximum estimates are in the companion's Section 7, not in the elementary bounded-increment argument above. Subcritical versions are given in [the slow-negative proof](#) and [the record-only extension](#), the latter for LCM numerators with the bound imposed only at late record steps. These results do not establish the general bound on the sum at record steps.

8 The remaining arithmetic estimate

Proposition 8.1 (necessary profile). *The integer tail of a sequence satisfying Problem 1.3's hypotheses but not its conclusion has $E_n \neq 0$ eventually, $|E_n|/C_n \rightarrow 0$, unbounded negative magnitudes along negative indices, and*

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

Proof. Absorption excludes late zeros, descent excludes an eventually nonnegative error, and Theorems 1.2 and 5.1 exclude the two finiteness conditions. $\square$

The profile is a necessary condition, not a construction. For a zero-indexed exact orbit with $a_n > 1$, $C_n > 0$, $D_n \geq 0$ and $E_n/C_n \rightarrow 0$, the companion's Section 4 proves that $D_n > 0$, $C_n/D_n \rightarrow 0$ and $\sum_n 1/a_n = C_0/D_0$; it also derives quadratic growth. Thus these global orbit assumptions already supply reciprocal-series realisation. A scalar numerical profile or a finite admissible prefix does not.

Problem 8.2 (growth of repeated denominator factors). For every rational-tail orbit satisfying the hypotheses but not the conclusion of Problem 1.3, must

$$\limsup_{n \rightarrow \infty} \frac{\log M_n}{n} > 0?$$

Equivalently, must there be a $K \geq 1$ for which

$$2^n \leq M_n^K$$

at infinitely many indices?

Section 4 gives $C_{n+1}/C_n \rightarrow 1$; averaging logarithms therefore gives $\log C_n = o(n)$. Since $1 \leq M_n \leq C_n$, every such orbit already satisfies $\log M_n/n \rightarrow 0$. The proposed positive upper limit is therefore a contradiction target: proving it from failure of Sylvester behaviour would exclude every counterexample, not describe an additional possible growth regime.

The necessary profile is proved above and is formalised as [the necessary conditions on a counterexample](#). The formal statement assumes the rational reciprocal sum, the quadratic growth limit and failure of eventual Sylvester behaviour; the last hypothesis is essential to all of its nontermination conclusions.

For $B \geq 0$ put

$$F_B(X) = \sum_{\substack{n \in \mathcal{R} \\ U_n \leq X}} (-V_n - B)_+.$$

For a rational tail satisfying the growth hypothesis, the weighted criterion reduces the original problem to proving that, for some fixed integer $B \geq 0$,

$$\liminf_{X \rightarrow \infty} \frac{F_B(X)}{X} = 0. \tag{5}$$

If the recurrence is not eventually Sylvester, first crossings instead give $F_B(X) \geq X/P_B - O_B(1)$, with an orbit-dependent CRT modulus P_B . The sum groups each record step by its starting numerator U_n , not by the running maximum R_n or the time index. On a non-Sylvester tail, $U_n \rightarrow \infty$, so only finitely many of these steps begin below any fixed X . The following lemma proves equivalence with the existence of an admissible weight. Enumerate the record indices $n \in \mathcal{R}$ by j , and apply it with $u_j = U_n$ and $w_j = (-V_n - B)_+$. The missing assertion is that growth and rationality force (5); the first-crossing lower bound is already proved.

Lemma 8.3 (weights and linear density). *Let u_j be positive integers, let $w_j \geq 0$, and put $F(X) = \sum_{u_j \leq X} w_j$, with the sum allowed a priori to be $+\infty$. Then $\liminf_{X \rightarrow \infty} F(X)/X = 0$ if and only if there is a finite nonincreasing $f : [1, \infty) \rightarrow [0, \infty)$ with $\int_1^\infty f(t) dt = \infty$ and $\sum_j w_j f(u_j) < \infty$.*

Proof. Suppose the lower limit vanishes. Choose cutoffs at which the ratio $F(X)/X$ is small. Since F is constant on each $[m, m+1)$, there are integers $X_1 < X_2 < \dots$ with $X_k \geq 2^k$ and $F(X_k) \leq 2^{-k} X_k$. Set

$$f(t) = \sum_{k \geq 1} \frac{\mathbf{1}_{[1, X_k]}(t)}{X_k}.$$

This function is nonincreasing and bounded by $\sum_k 2^{-k}$. Each cutoff contributes $1 - 1/X_k$ to its integral, but at most 2^{-k} to the weighted sum. Interchanging nonnegative sums therefore gives

$$\int_1^\infty f(t) dt = \sum_k (1 - 1/X_k) = \infty, \quad \sum_j w_j f(u_j) = \sum_k \frac{F(X_k)}{X_k} \leq 1.$$

Conversely, let f be such a weight and suppose $F(m) \geq cm$ for some $c > 0$ and all integers $m \geq M$. A nonincreasing nonnegative function with divergent integral is positive everywhere, so $F(X) \leq f(X)^{-1} \sum_j w_j f(u_j)$ is finite. Partial summation gives, for $N > M$,

$$\sum_{u_j \leq N} w_j f(u_j) = F(N)f(N) + \sum_{m=1}^{N-1} F(m)(f(m) - f(m+1)) \geq c(Mf(M) + \sum_{m=M+1}^N f(m)),$$

every term being nonnegative. The right side diverges as $N \rightarrow \infty$, because $\sum_{m \geq 1} f(m) \geq \int_1^\infty f(t) dt = \infty$. $\square$

The remaining estimate must use arithmetic compatibility, not just the size of the error. Section 14 of the companion gives the scalar profile $C_n = n^2 + 1$, $E_n = -(2n + 1)$ and its failed divisibility check, as well as sequences avoiding sparse fixed moduli. Its discussion of reduced numerators also distinguishes compatibility modulo one old denominator from control of prime divisors of later denominators. None of those weaker properties supplies (5).

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A Sources and related methods

The mathematical arguments above are independent of the source code. Each linked phrase below opens its declaration at an immutable revision; the checkpoint is 3d6d938d696f unless the destination identifies a separate release or older research note.

Recorded coverage. The supplied index records checked proofs of the bounded-negative theorem, its product and LCM consequences, the scalar convergence criterion and necessary conditions on a counterexample. The rational-tail construction has its own formalised estimates; it is not merely an interpretation of an abstract recurrence. The [weighted criterion is checked in both directions](#), with the real cutoff used in Lemma 8.3. The cubic and double-logarithmic arguments have checked components, not assembled Lean proofs identified here; the companion supplies their written proofs, including the growth and modulus-counting arguments.

The recorded main build is at 6b78209ab63a; links retain their original pins, including 3d6d938d696f. The linked weighted criterion also has a fresh Lean check; this does not certify the remaining exposition as a whole. Finite certificates and unsuccessful extensions are retained in the companion, not counted as proofs of assembled arguments. The earlier Isabelle/HOL development of Koutsoukou-Argyriaki and Li [16] verifies older irrationality criteria, not Problem #243. None of these records supplies the missing increment bound or global record estimate.

Earlier remainder methods and related problems. The integer-remainder method predates this argument [12, Theorem 2.1, pp. 85–86]. Hančl and Tijdeman characterise rationality of polynomial Cantor series by a finite decomposition and a shifted polynomial sum [13, Theorem 2.2 and its derivation, pp. 39–40]; the polynomial hypotheses justify that rearrangement. Neither comparison supplies the lower-bound CRT argument. The freely selectable series of Problem #270 [15, Theorems 1–2] are discussed in the companion’s Section 3, but are not used for the fixed recurrence here.

Main criterion and arithmetic lemmas. The formalisation checks [the reciprocal-series corollary and its integer-tail construction](#) (Corollary 1.1). The proof of [eventual vanishing under a lower error bound](#) (Theorem 1.2) shifts past the two thresholds and applies the [vanishing result under a bound on the negative part](#) with centring and the lower bound at every index.

Proposition 3.1 corresponds to the [update law](#) and the [defect identity](#). Theorem 3.2 uses [absorption of a vanishing error](#) and [stabilisation of the nonnegative error](#). The [local rigidity step](#) and the [eventual Sylvester recurrence](#) give the two forms of Corollary 3.3.

The three parts of Proposition 3.6 are [step coprimality](#), [pairwise coprimality](#), and [coprimality to each earlier multiplier](#). Proposition 3.7 gives [eventual constancy of the gcd](#). The exclusions treat [reduced numerators with bounded increments](#) and [the case where](#)

divergence follows from vanishing relative error. The main proof uses both divergence of the reduced numerator and a bound on every upward increment. A cofinal bound on negative errors suffices for gcd stabilisation, not for that increment bound. Companion Section 11 also distinguishes stabilisation from a density-zero bound on strict gcd increases; the latter is not used here.

The scalar criterion. The scalar declaration proves [vanishing for a positive integer sequence with summable relative increases](#) (Theorem 5.1); its exact-orbit consequence is [the Sylvester recurrence under the same assumption](#). These declarations concern integer C_n , not the real-initial-value observation following the proof. Formulations that also assume denominator dynamics and vanishing relative error give [a growth bound for the numerator, eventual zero error under the convergence assumption](#), and [the resulting Sylvester recurrence](#).

The state dynamics and arithmetic exclusions are in `ReciprocalTailRigidity.lean`; the scalar criterion is in `SparseResetRecovery.lean`. The product and least-common-multiple criteria and the necessary conditions on a counterexample are in `PaperCompleteR7`. The checked periodic theorem assumes that each error magnitude is smaller than its multiplier. For periodic positive magnitudes and multipliers $a_n \geq 2$, the companion derives this bound eventually from the exact recurrences. For general errors, vanishing relative error supplies strict centring eventually, but control of the negative part is still needed in Theorem 1.2.

B A factorial residue reduction for forced orbits

A constant-negative error $E_n = -m$ with $C_0 = c$ produces $C_n = c + nm$ and the shape equation

$$D_n + m = (a_n - 1)C_n. \quad (6)$$

In the case $m = c = 1$, where (6) reads $D_n + 1 = (a_n - 1)(n + 1)$, each multiplier is determined by its predecessor; we call such an orbit *forced*. At index n the numerator of the next multiplier is

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the [forced numerator](#), and the divisor is $n + 2$. The orbit continues for another step exactly when this division is exact. Here survival counts exact divisions; it does not impose the positivity conditions of an infinite reciprocal-tail orbit. The [formal survival predicate](#) requires exact division at each step.

Example B.1 (the shape equation running until it fails). The forced orbit from $a = 3$ reads this way: $\text{num}(0, 3) = 6$ is divisible by 2 and gives $a_1 = 3$, while $\text{num}(1, 3) = 13$ is not divisible by 3, so the orbit stops there.

Direct iteration can produce very large intermediate integers. To decide whether the first h divisions are exact, however, it suffices to know the initial value modulo $(h + 1)!$. Computing a pseudo-greedy orbit through residues modulo a shrinking product modulus is Koizumi's method [10, Remark 2 and Algorithm 1, pp. 13–14]; for the forced numerator that product is a factorial.

Theorem B.2 (factorial residue reduction). *For all h and all integers $a \equiv b \pmod{(h+1)!}$, the orbit from a survives h forced updates exactly when the orbit from b does.*

Proof. Define $M(0, i) = 1$ and $M(h+1, i) = (i+2)M(h, i+1)$. Thus $M(h, 0) = (h+1)!$. We prove the stronger statement that, at index i , congruent inputs modulo $M(h, i)$ survive the same h updates. There is nothing to prove for $h = 0$.

For the inductive step, suppose $a \equiv b \pmod{(i+2)M(h, i+1)}$. Since $\text{num}(i, \cdot)$ is an integer polynomial, its values at a and b are congruent modulo that product. In particular, one is divisible by $i+2$ if and only if the other is. If neither is divisible, both orbits stop. Otherwise their quotients are congruent modulo $M(h, i+1)$, so the induction hypothesis applies to the remaining h updates at index $i+1$. Taking $i = 0$ proves the claim. $\square$

The formal [factorial residue reduction](#) uses the [shrinking-modulus induction](#), [polynomial congruence](#), and [cancellation after exact division](#). Its modulus is identified by the [ascending-factorial formula](#) and its [factorial value at the initial index](#).

At $h = 1$, survival means $2 \mid a^2 - 2a + 3$, or equivalently that a is odd. This illustrates Theorem B.2 with modulus $2! = 2$.

Exact enumeration over $2 \leq a_0 < 5000$ gives at most 17 successful updates (18 values including a_0). The omitted seed $a_0 = 1$ is a fixed point, outside $a_n \geq 2$. This finite search is not a proof of the [constant-negative exclusion](#), which concerns infinite natural-number orbits under its stated hypotheses. Example 9.1 of the companion gives the further search details.

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